

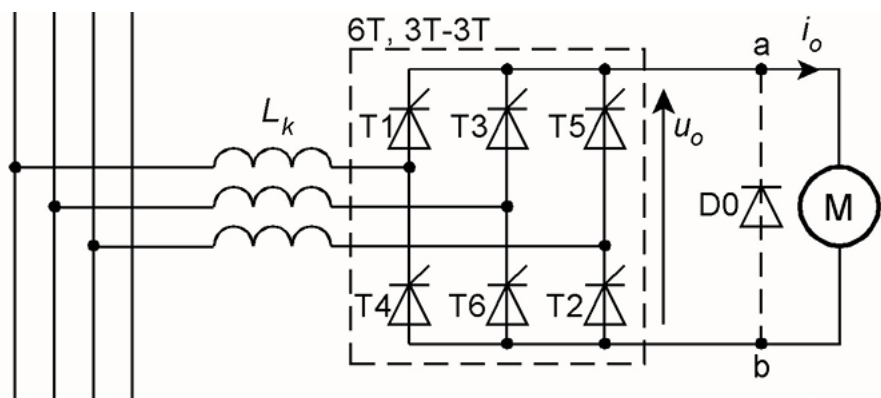
6T Thyristor Rectifier Simulator

MATLAB/Simulink project for simulation and analysis of a three-phase fully controlled 6-pulse thyristor rectifier with an R-L-E load.

The project demonstrates both rectifier and line-commutated inverter operation, automatic operating point calculation, DC-side waveform analysis, validation scripts, and a MATLAB-based graphical control panel.

Features include:

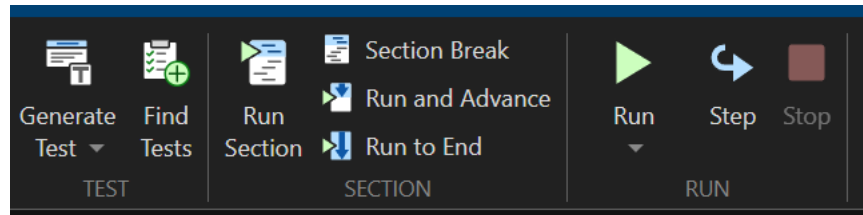
- 6T fully controlled thyristor bridge model
- rectifier and line-commutated inverter operation
- automatic α / E_o operating point calculation
- MATLAB graphical control panel
- waveform plotting and validation results



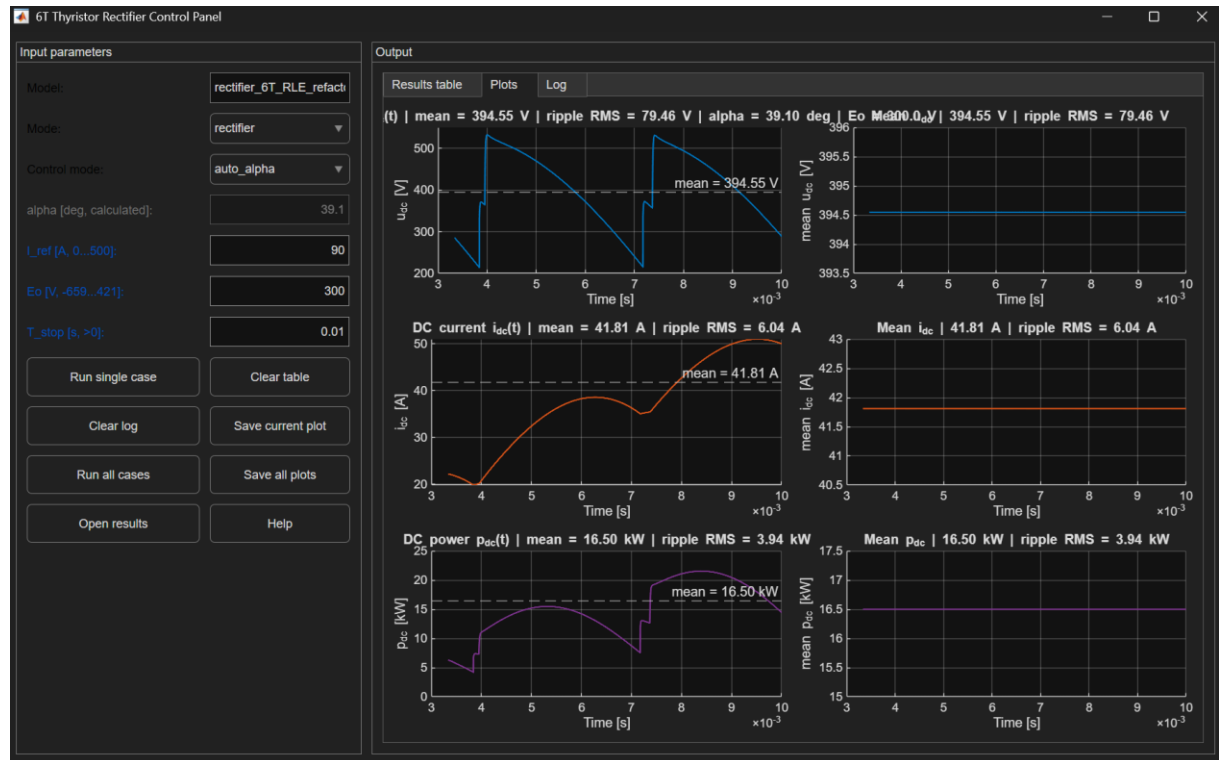
How to access the panel:

1. Open the 'rectifier_control_panel' matlab file. Make sure that the other .m and .slx files are in the same folder.

2. Press „Run” on the top of the program



3. After a while a control panel will be visible.



4. After selecting the modes and inputting the values, press Run single case to display the result table and the plots.

5. Typical validation results for the rectifier can be downloaded by pressing ‘Run all cases’.

Theory

The thyristors do not conduct immediately when they become forward biased. They start conducting only after receiving a gate pulse. The delay between the natural commutation instant and the gate pulse is called the firing angle defined as α . Increasing the firing angle delays the conduction of each thyristor pair. As a result, the average DC output voltage decreases. For a fully controlled three-phase bridge operating with continuous DC current, the average output voltage can be approximated as:

$$U_d = U_{d0} \cos(\alpha)$$

where:

$$U_{d0} = 1.35 U_{LL}$$

For a 400 V line-to-line RMS supply:

$$U_{d0} = 1.35 \cdot 400 \text{ V} = 540 \text{ V}$$

Thus, in the ideal case:

alpha = 0 deg -> maximum positive DC voltage

alpha = 90 deg -> average DC voltage close to zero

alpha > 90 deg -> negative average DC voltage

This relationship is the basis for controlling the rectifier output voltage.